Week 3 - Part 1 - Osprey1st College Algebra - Summer 2024

Section covered: 2.2 and 2.3

Recall that a term is a number or the product of a number and variables raised to powers. The terms of the expression $4x^2 - 3x$ are $4x^2$ and $-3x$.

Terms and Coefficients of Expression

degree: 5

Expression	
$4x^5 - x^3 + 7x^2 + \frac{x}{2} - 5$	
Terms	**Coefficients**
$4x^5$	4, leading coefficient
$-x^3$	-1
$7x^2$	7
$\frac{x}{2} = \frac{1}{2}$ x	$\frac{1}{2}$
-5	-5 constant

Polynomial
A polynomial in x is a finite sum of terms of the form ax^n, where a is a real number and n is a **whole number**.

The degree of a polynomial is the greatest exponent of x among all terms. The degree of a constant is 0.

A **monomial** has one term, a **binomial** has two terms, and a **trinomial** has three terms.

Problem 1. Determine whether the expression is a polynomial. If it is a polynomial, find the degree and the leading coefficient.

(a) * $-7v^4 \boxed{+18v^6} - 1 - 23v$ degree : 6 LC: 18

Yes

(b) $3x^{-2} + 8x + 1$ $\frac{3}{x^2}$

No

(c) $4x^3 + \dfrac{6}{x^4} - 8$ No

Problem 2. Add or subtract the polynomials.

(a) * $(-2y^2 + y + 4) + (4y^2 + 2y - 6) - 1(7y^2 + 3y - 1)$ $\boxed{-5y^2 - 1}$

$-7y^2 - 3y + 1$

(b) $(2x^3 - x + 1) - 4(x^3 - x^2 - 2x + 5)$ $\boxed{-2x^3 + 4x^2 - 9x - 19}$

$-4x^3 + 4x^2 - 8x - 20$

Problem 3. Multiply polynomials.

(a) * $-7x^3(-3x^3 + 7x - 4)$ $= 21x^6 - 49x^4 + 28x^3$

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(b) * $(5c + 4)(5c - 6)$

$25c^2 - 30c$
$+20c - 24$
$25c^2 - 10c - 24$

(c) $(4z - 3)(2z^2 - z + 5)$

$8z^3 - 4z^2 + 20z$
$-6z^2 + 3z - 15$
$8z^3 - 10z^2 + 23z - 15$

(d) * $(7u^2 - 3u - 5)(6u^2 - 2u - 5)$

(d) $42u^4 - 14u^3 - 35u^2$
$-18u^3 + 6u^2 - 15u$
$-30u^2 + 10u + 25$
$42u^4 - 32u^3 - 59u^2 - 15u + 25$

(e) $(3x - 2)(3x + 2)$

$9x^2 + \cancel{6x}$
$\cancel{-6x} - 4$
$9x^2 - 4$

(f) $(x + 8)^2$

$(x+8)(x+8)$

$x^2 + 8x$
$+8x + 64$
$x^2 + 16x + 64$

$2(8x)$
$x^2 + 16x + 64$

Special Products

Product of the Sum and Difference of Two Terms $(A + B)(A - B) = A^2 - B^2$

Square of Binomials $(A + B)^2 = A^2 + 2AB + B^2$

$(A - B)^2 = A^2 - 2AB + B^2$

Problem 4. Multiply the polynomials.

(a) $(x+2)(x-2)$

$x^2 - 4$

(b) $(7a-9)(7a+9)$

$49a^2 - 81$

(c) * $(7y-5x)(7y+5x)$

$49y^2 - 25x^2$

(d) $(5x^2-8y^4)(5x^2+8y^4)$

$25x^4 - 64y^8$

(e) $x^3(2x-5)(2x+5)$

$x^3(4x^2-25) = 4x^5 - 25x^3$

(f) $(8z+3)^2$ $\quad 2(8z)3$

$64z^2 + 48z + 9$

(g) $(7x-4)^2$ $\quad 2(7x)4$ $\qquad$ FOIL $(7x-4)(7x-4)$

$49x^2 - 56x + 16$

(h) * $(2-5w)^2$ $\quad 2(2)5w$

$4 - 20w + 25w^2$

(i) $(s+t)^3$ $\quad (s+t)(s+t)^2$

$(s+t)(s^2+2st+t^2)$

$s^3 + 2s^2t + st^2$

$s^2t + 2st^2 + t^3$

$s^3 + 3s^2t + 3st^2 + t^3$

Factors and Factoring

Factoring is the reverse process of multiplying.

Product	**Factored Form**
18	$9 \cdot 2$
$12x^3 + 6$	$6 \cdot (2x^3 + 1)$
$9x^2 + 3x$	$3x \cdot (3x + 1)$
$-24y + 36$	$-12 \cdot (2y - 3)$

The Greatest Common Factor (GCF) of a list of terms contains the smallest exponent on each common variable.

$$\text{The GCF of} \quad x^5y^6, \quad x^2y^7, \quad \text{and} \quad x^3y^4 \quad \text{is} \quad x^2y^4.$$

Problem 5. Find the greatest common factor.

(a) 35 and 42 $= 7$

(b) 12y, 18y, and 36y $= 6y$

(c) x^2 and x^5 $= x^2$

(d) $4x^2$, $8x^3$, and $14x^4$ $= 2x^2$

(e) * $45u^5, 63u^3$ and $27u^2$ $= 9u^2$

(f) * $14v^4u^5y^6$ and $7v^2y^3$ $= 7v^2y^3$

$\frac{45}{3}$ $\quad \frac{63}{3}$ $\quad \frac{27}{3}$

9 $\quad \frac{15}{3}$ $\quad \frac{21}{3}$ $\quad \frac{9}{3}$

5 $\quad$ 7 $\quad$ 3

Problem 6. Factor the following expression (factor out the greatest common factor).

(a) $\frac{30x}{10} - \frac{20}{10} = 10(3x - 2)$

(b) $\frac{6x^2}{6x^2} - \frac{18x^4}{6x^2} = 6x^2(1 - 3x^2)$

(c) * $20n^2 + 15n^3 = 5n^2(4 + 3n)$

(d) $8x^4 - 12x^3 + 20x^2 = 4x^2(2x^2 - 3x + 5)$

(e) * $\frac{18uw^2y^5}{6uw^2} + \frac{30u^6w^8}{6uw^2} = 6uw^2(3y^5 + 5u^5w^6)$

Problem 7. Factor the following expression.

(a) $x(4x+1)+5(4x+1) = (4x+1)(x+5)$

(b) * $5(x+9)-3x(x+9) = (x+9)(5-3x)$

(c) $1(2a+5)-b(2a+5) = (2a+5)(1-b)$

~~(d) $24(y-2)^3-16(y-2)-y+2$~~

Factor by Grouping

$$
\begin{aligned}
&xy+2x+3y+6 && \\
&=(xy+2x)+(3y+6) && \text{Group terms} \\
&=x(y+2)+3(y+2) && \text{Factor out in each grouping} \\
&=(y+2)(x+3) && \text{Factor by Grouping}
\end{aligned}
$$

Problem 8. Factor by grouping.

(a) $(15x^3-10x^2)(+6x-4)$

$5x^2(3x-2)+2(3x-2)$

$(3x-2)(5x^2+2)$

(b) * $(y^3+7y^2)(+5y+35)$

$y^2(y+7)+5(y+7)$

$(y+7)(y^2+5)$

(c) $3xy-3ay-6ax+6a^2$

(d) * $(uy+7y)(-7u-49)$

$y(u+7)-7(u+7)$

$(u+7)(y-7)$

(e) $(r^2s+3r^2)(-5s-15)$

$r^2(s+3)-5(s+3)$

$(s+3)(r^2-5)$

(f) $9y^3-15y^2+3y-5$